Fisher Information Loss for Coordination Rules in Decentralized Sensor-Agent Networks

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Abstract

Decentralized sensor networks routinely face a last-mile coordination problem: each mote or edge agent observes only a local reading and a local coordinate, yet the team must decide *who exchanges what with whom*, and when to *quarantine* a report under uncertainty. Perception—local categorical distributions induced by readings and geometry—and coordination—a deterministic rule that compresses the joint into exchange or quarantine actions—are often designed separately. While Dec-POMDP active perception and information-driven sensing carefully treat what to observe, the coordination layer is frequently heuristic and evaluated only by task reward or throughput. This paper asks a complementary question: *holding the agents' local laws fixed, which θ-independent coordination rules destroy the least Fisher information about the global field or event parameters?*

We formalize coordination as an AggregationMap $A = f(Z)$ and quantify information loss by the data-processing gap $I_Z(\theta) - I_A(\theta) \succeq 0$ between the Fisher information of a declared product joint and that of its pushforward. Rules are ranked by $\mathrm{tr}(\Delta) = \mathrm{tr}(I_Z - I_A)$. The intended primary application is decentralized sensor networks. On a synthetic unit-disk 2D field (9 motes) the core order holds and quarantine-wider is the lowest-loss non-trivial rule; on a 12-mote Intel Berkeley Lab (IBRL) real-data control the core order also holds, but both quarantine rules lose more than baseline, so the quarantine safety valve pays an information cost and the full ranking is data-dependent. On the IMAS^2 grid-world benchmark with fixed sensors, the Fisher-loss ranking of fusion rules tracks conditional entropy and accuracy, while Fisher ΔI proves a weak sensor-placement criterion. A no-recalibration cross-bed transfer of AggregationMap rankings fails Heath's tongyong and zhengxiang criteria on most of the six directed pairs (Section 8), so the evidence supports within-bed auditing only. As a reproducible analytic illustration of the same AggregationMap programme, we also use the Jevometry three-agent data-cleaning council (applicability, repair, risk) and compare seven pure rules—baseline, stricter, wider, quarantine-to-review, a hybrid quarantine-wider split, coarse binary, and a near-identity upper bound—on a grid over evidence and ambiguity. At the design centre, baseline coordination discards roughly half of $\mathrm{tr}(I_Z)$; softening quarantine into human review recovers a substantial fraction of that loss, and combining that softening with distinct auto-normalize/auto-impute labels yields a further improvement that holds on the full design grid. The ordering near-identity $\leq$ wider $\leq$ baseline $\leq$ stricter is stable on every grid cell. We discuss design implications for sensor-agent exchange and quarantine policies, API constraints that force deterministic maps, and the role of the cleaning numbers as an analytic illustration alongside the unit-disk and IBRL results. Accompanying code and reproducible tables ship with the Jevometry repository.

1 Introduction

Modern sensing stacks make it cheap for each mote or edge agent to emit a calibrated-looking categorical distribution over local judgments: observation quality, whether to report, which neighbour to exchange with, or whether a reading should be quarantined. In a decentralized

sensor network these local judgments are rarely the end product. A coordinator—often a short piece of deterministic logic shared by the team—compresses the joint of agent outputs into exchange decisions (who/what to exchange) or a quarantine safety valve [Jevometry maintainers, 2026].

This compression is an information bottleneck. Classical decision theory and Dec-POMDP planning optimize rewards defined on beliefs or on prediction quality [Lauri and Oliehoek, 2020, Lauri et al., 2020, Oliehoek and Amato, 2016]. Information-theoretic multi-agent sensing maximizes mutual information or related set functions, often with submodular greedy guarantees [Krause et al., 2008, Nemhauser et al., 1978, Corah and Michael, 2021, Shi et al., 2026]. Those lines of work choose *what to observe* or *who should observe*. They do not, by themselves, tell a practitioner which of two already-deployed coordination rules retains more sensitivity to the global field or event parameters θ the network was designed to track.

Fisher information provides a local, differentiable language for that sensitivity [Fisher, 1925, Amari, 2016]. For a discrete model $P_\theta(Z)$ with score Jacobian J_Z, the Fisher matrix $I_Z = J_Z^\top \mathrm{diag}(P_\theta)^{-1} J_Z$ measures how strongly infinitesimal parameter moves change the outcome distribution. When a fixed map f produces actions $A = f(Z)$, the data-processing inequality for Fisher information asserts $I_Z - I_A \succeq 0$ under standard regularity conditions [Zamir, 1998, Cover and Thomas, 2006]. The scalar $\mathrm{tr}(I_Z - I_A)$ is then a comparable loss summary when I_Z is held fixed.

Target setting and current evidence. Our primary narrative is decentralized sensor agents under local observations, using Fisher ΔI as a shared decision function to choose among coordination rules—including quarantine when uncertain. The intended main experiment is a synthetic unit-disk field in 2D; a secondary control is a 12-mote Intel Berkeley Lab (IBRL) subnet. Both are reported in Section 6.5. We additionally report a reproducible analytic instantiation of the same AggregationMap programme: the Jevometry three-agent cleaning council, whose tables and figures serve as a proof of concept for ranking `near`/`wider`/`baseline`/`stricter`/`quarantine_*`/`near_identity`-style rules by $\mathrm{tr}(\Delta)$.

Contributions.

1. We cast multi-agent coordination-rule comparison as ranking θ-independent aggregation maps by Fisher information loss under a shared declared joint, with sensor-network exchange and quarantine as the motivating AggregationMap semantics (Section 3).

2. We map the programme onto three layers—global field parameters θ, local agent categorical laws, and θ-independent AggregationMaps—and run a synthetic unit-disk main experiment and an IBRL real-data control (Sections 4 and 6.5). The core order holds on both; quarantine-wider is best on unit-disk (1.784), whereas on IBRL both quarantine rules lose more than baseline (3.042, 3.101 vs. 2.785).

3. On the IMAS^2 grid world with fixed sensors, the $\mathrm{tr}\Delta I$ ranking of fusion rules has Spearman 0.65–0.93 with conditional entropy and 0.51–0.87 with negative accuracy; as a sensor-placement criterion Fisher ΔI is weak and fails under stochastic motion, a negative finding we report as such (Section 7).

4. A no-recalibration cross-bed transfer among unit-disk, IBRL, and IMAS^2 fails as a boundary: tongyong 2/6, zhengxiang 0/6, S1 0/6, S2 3/6; priority directions unit-disk$\to$IMAS^2 and IMAS^2 $\to$IBRL both fail (Section 8).

5. As a current analytic instantiation, we instantiate seven pure rules in the Jevometry cleaning council and provide a reproducible harness that sweeps a parameter grid, exports CSV/JSON (including parameter-axis loss diagonals and fractional losses), and emits publication figures (Section 5).

6. Empirically, at the design centre the baseline rule loses about 44% of $\mathrm{tr}(I_Z)$; quarantine-to-review cuts loss by roughly a quarter relative to baseline; a hybrid quarantine-wider rule improves further ($\mathrm{tr}(\Delta) \approx 2.16$) and dominates quarantine-to-review on the full grid; a binary gate loses more than stricter mid-risk rejection (Section 6). These numbers illustrate the AggregationMap audit analytically; sensor results are in Section 6.5.

7. We discuss how this metric complements reward-centric Dec-POMDP evaluation for sensor coordination and what API constraints (deterministic maps only) imply for future learned coordinators (Section 9).

This manuscript is intended as a short journal / arXiv draft accompanying open-source experiments. Cleaning-council numbers are analytic and illustrative; the unit-disk and IBRL sensor results are the primary empirical story.

2 Related work

Dec-POMDP active perception. Lauri and Oliehoek [2020] reduce multi-agent active perception with prediction rewards to standard Dec-POMDP solvers via individual prediction actions, with decentralization-loss bounds. Lauri et al. [2020] optimize convex belief-dependent rewards (e.g., negative entropy) with policy graph improvement in discrete and continuous-state settings. Both lines typically treat the team as fixed and fold information objectives into rewards rather than post-hoc comparing deterministic coordinators.

Information-driven multi-agent sensing. Submodular sensor selection and multi-robot tracking maximize mutual information or related coverage objectives with greedy $(1 - 1/e)$ guarantees under monotonicity [Krause et al., 2008, Corah and Michael, 2021]. Olkin et al. [2024] and Molloy and Nair [2023] further connect planning to information and smoother entropy.

IMAS^2 (closest Dec-POMDP relative). Shi et al. [2026] study *joint* sensing-agent selection and decentralized active-perception policy synthesis in Dec-POMDPs (`arXiv:2510.20009`; AAMAS 2026). An inner layer maximizes mutual information between latent trajectories (or a secret property) and the team's partial observations; an outer layer greedily selects agents under submodularity, with a $(1 - 1/e)$-style guarantee when successive marginal gains satisfy their stated ratio condition. Experiments use multi-agent cooperative perception in a grid world. That programme *optimizes who senses and how* under Dec-POMDP dynamics. Ours is a related but distinct route: we *hold* the local agent laws fixed and rank θ-independent AggregationMaps by Fisher information loss $\Delta_f = I_Z - I_A$ after action aggregation—including quarantine as a deterministic safety valve—rather than maximizing mutual information via submodular agent selection. The two metrics answer complementary questions (select/synthesize sensors vs. audit last-mile coordination).

Information geometry for agents. Fisher–Rao geometry supplies local metrics on parametric families of distributions [Amari, 2016]. Jevometry operationalizes node- and system-level Fisher diagnostics for agent systems [Jevometry maintainers, 2026]. We use its pushforward `information_loss` primitive and do not modify the underlying DECLARED_PRODUCT construction.

3 Problem formulation

3.1 Declared product joint

Let $\theta \in \Theta \subset \mathbb{R}^d$ parameterize a multi-agent system. Each agent $i \in \{1, \ldots, m\}$ emits a categorical distribution $P_\theta(Z_i)$ over a finite local alphabet $\mathcal{Z}_i$. A *declared product joint* asserts conditional independence given θ:

$$P_\theta(Z = z) = \prod_{i=1}^{m} P_\theta(Z_i = z_i), \qquad z = (z_1, \ldots, z_m) \in \mathcal{Z} = \prod_i \mathcal{Z}_i. \tag{1}$$

In the cleaning instantiation below, $m = 3$, $d = 2$, and $\theta = (\text{evidence}, \text{ambiguity})$ with coordinates in $[0, 1]$. In the planned sensor instantiation, θ instead denotes global field or event parameters (e.g., source location, intensity, or event indicators), not raw single-node readings. The product declaration is an assumption of the model, not an empirical claim about live APIs or radio traces.

3.2 Coordination as aggregation

Definition 1 (Coordination rule)**.** *A coordination rule is a map $f : \mathcal{Z} \to \mathcal{A}$ into a finite action alphabet $\mathcal{A}$. We call f* pure *if it is a total function of the joint outcome alone (no hidden randomness) and θ-*independent *if f does not depend on θ.*

The induced action law is the pushforward

$$P_\theta(A = a) = \sum_{z : f(z) = a} P_\theta(Z = z). \tag{2}$$

If $J_Z(\theta)$ is the Jacobian of the joint probability vector with respect to θ, the aggregate Jacobian J_A is obtained by summing rows of J_Z that share an action label.

3.3 Fisher information loss

Definition 2 (Information loss of a rule)**.** *For a θ-independent rule f, define*

$$I_Z(\theta) = J_Z(\theta)^\top \operatorname{diag}\big(P_\theta(Z)\big)^{-1} J_Z(\theta), \tag{3}$$

$$I_A(\theta) = J_A(\theta)^\top \operatorname{diag}\big(P_\theta(A)\big)^{-1} J_A(\theta), \tag{4}$$

$$\Delta_f(\theta) = I_Z(\theta) - I_A(\theta). \tag{5}$$

We report $\operatorname{tr}(I_Z)$, $\operatorname{tr}(I_A)$, and $\operatorname{tr}(\Delta_f)$ as scalar summaries, and we verify $\Delta_f \succeq 0$ numerically (PSD check) as a data-processing diagnostic.

Proposition 1 (DPI for fixed maps)**.** *Under standard positivity and differentiability conditions, any θ-independent f satisfies $\Delta_f(\theta) \succeq 0$. Equality holds when f is injective on the support of P_θ.*

Remark 1. *Maps that depend on θ (soft policies, learned routers conditioned on parameters) fall outside this guarantee in the Jevometry implementation and are reported as unsupported. This paper therefore restricts attention to pure, fully covering, θ-independent rules—the class that matches typical production if-then coordinators, mote firmware exchange policies, and the library's AggregationMap contract.*

Comparison principle. Fix the agents and the joint construction. Rank rules by $\operatorname{tr}(\Delta_f)$ (lower is better) or equivalently by $\operatorname{tr}(I_A)$ (higher is better). Because I_Z is shared, ranking by loss is identical to ranking by retained information.

4 Three layers for sensor-agent coordination

The AggregationMap programme admits a three-layer reading that matches decentralized sensing:

(i) **Global parameters θ.** θ denotes field- or event-level quantities of interest (source location, intensity, spatial covariance hyperparameters, binary event indicators), *not* the raw scalar reading at a single mote. Fisher loss is measured with respect to this θ.

(ii) **Local agent laws.** Each sensor is an agent. Its categorical distribution $P_\theta(Z_i)$ is induced by local coordinates and local readings—for example, bins of observation quality, a report/hold decision, or a neighbour-choice alphabet under a unit-disk communication graph. The declared product joint assembles these local laws.

(iii) **AggregationMap as exchange and quarantine.** A θ-independent rule f compresses the joint into coordination actions: who exchanges with whom, what payload to send, or whether to quarantine a report when local uncertainty is high. Candidate families reuse the same qualitative knobs studied below—near-neighbour exchange, wider neighbourhoods, baseline firmware, stricter thresholds, quarantine variants, and a near-identity upper bound—and are ranked by $\mathrm{tr}(\Delta_f) = \mathrm{tr}(I_Z - I_A)$.

Sensor experiments.

- **Main:** synthetic unit-disk sensors in a 2D field (9 motes, 3×3 grid, $R = 0.45$). Local alphabets and AggregationMaps follow layers (ii)–(iii); results are in Table 2.
- **Control:** Intel Berkeley Lab (IBRL) subnet of 12 motes (ids 4, 11, 16, 18, 20, 26, 34, 39, 42, 47, 50, 53), as a real-deployment sanity check under the same AggregationMap interface; results are in Table 3.

The cleaning-council study in Sections 5–6 is the present, fully reproducible analytic illustration of layers (i)–(iii) with $\theta = (\text{evidence}, \text{ambiguity})$ and repair/quarantine actions standing in for exchange/quarantine semantics.

5 Method and experimental design

5.1 Three-agent cleaning council (analytic instantiation)

We reuse the offline Jevometry tutorial `examples/three_agent_cleaning` [Jevometry maintainers, 2026] as a proof-of-concept AggregationMap audit—not as a claim about live sensor fields. Three analytic softmax agents produce distributions for a single candidate repair of a noisy numeric field:

Agent	Primitive	Support
applicability	Noul	$\{\text{false}, \text{true}\}$
repair	Choice	$\{\text{normalize}, \text{impute}, \text{quarantine}\}$
risk	Score	$\{0, 1, 2\}$ (low/med/high)

Local logits are affine in $(\theta - \theta_0)$ around the design centre $\theta_0 = (0.7, 0.3)$, and probabilities are obtained by softmax. The joint is the declared product (ConstructionMode.DECLARED_PRODUCT). No live API is queried; no actual database write is performed. Quarantine in this council plays the same structural role as a sensor quarantine valve: a coarse action that merges distinguishable joint outcomes and therefore inflates $\mathrm{tr}(\Delta_f)$ unless softened (e.g., into human review or delayed exchange).

5.2 Coordination rules

We compare seven rules, all pure and θ-independent:

baseline Reject if inapplicable, quarantine, or high risk; auto-fix if normalize and low risk; otherwise human review.

stricter As baseline, but medium risk also rejects.

wider As baseline rejects; low-risk normalize and impute become distinct auto labels (`auto_normalize`, `auto_impute`).

quarantine_review Quarantine routes to human review instead of reject; reject only on inapplicable or high risk.

quarantine_wider Quarantine-to-review combined with wider's dual auto labels: the empirically strongest coarse policy in this study.

coarse_binary Reject under baseline reject conditions; otherwise a single `proceed` action (auto-apply and review collapsed).

near_identity One action label per joint outcome tuple (18 labels): Fisher retention upper bound under this alphabet.

In sensor language, baseline/stricter/wider correspond to narrower or broader exchange policies and risk thresholds; quarantine variants are safety valves; near-identity is an information-retention upper bound rather than a deployable radio schedule.

5.3 Harness

The script `paper_experiments.py` builds nodes and joint once, evaluates I_Z, I_A, and Δ_f on an 8×8 grid over evidence $\in [0.2, 0.9]$ and ambiguity $\in [0.1, 0.8]$, and additionally records the exact centre θ_0. Outputs:

- `papers/coord-fisher/data/rule_loss_grid.{csv,json}`
- `papers/coord-fisher/data/center_summary.json`
- `papers/coord-fisher/data/center_loss_anisotropy.csv`, `rule_loss_anisotropy_grid.csv`, fractional-loss heatmaps
- `papers/coord-fisher/data/iteration_summary.json`
- Figures 1–4 under `papers/coord-fisher/figures/`

Library code under `src/jevometry` is unchanged; only example-level rules and scripts are extended. The unit-disk and IBRL harnesses reuse the same AggregationMap and `information_loss` interface.

6 Experiments

6.1 Centre-point ranking

Table 1 reports traces at θ_0. All rules return status `ok` with PSD loss matrices. The near-identity map loses nothing (up to numerical noise), confirming the implementation against the equality case of the DPI. Baseline loses $\mathrm{tr}(\Delta) \approx 3.03$ out of $\mathrm{tr}(I_Z) \approx 6.84$—about 44%. Wider slightly improves on baseline by preserving distinct auto paths. Stricter mid-risk rejection increases

Table 1: Fisher traces at centre $\theta_0 = (0.7, 0.3)$, sorted by ascending loss. $|A|$ is the number of actions with positive mass under the pushforward. Cleaning-council analytic illustration—not unit-disk or IBRL.

Rule	$\mathrm{tr}(I_Z)$	$\mathrm{tr}(I_A)$	$\mathrm{tr}(\Delta)$	$\lvert A\rvert$	Status
near_identity	6.8418	6.8418	0.0000	18	ok
quarantine_wider	6.8418	4.6816	2.1602	4	ok
quarantine_review	6.8418	4.5764	2.2654	3	ok
wider	6.8418	3.8285	3.0133	4	ok
baseline	6.8418	3.8080	3.0338	3	ok
stricter	6.8418	3.4198	3.4220	3	ok
coarse_binary	6.8418	3.1296	3.7122	2	ok

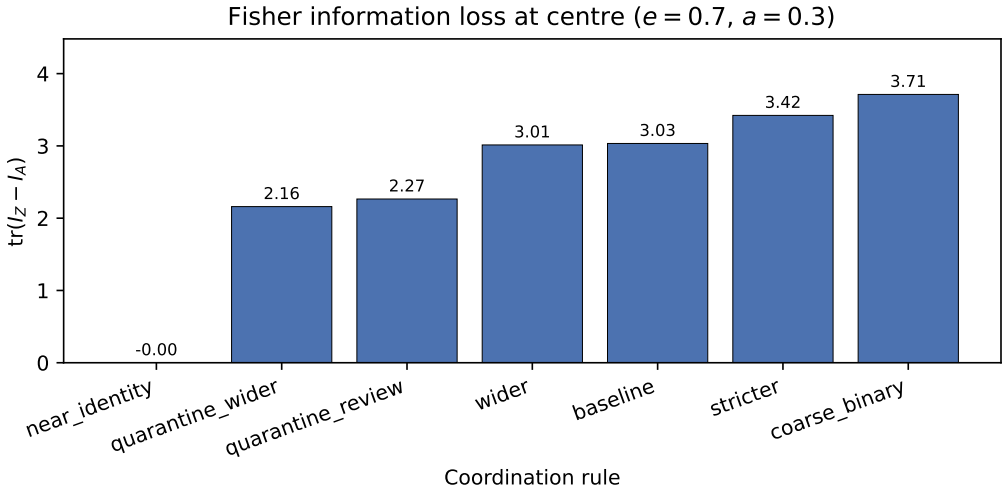


Figure 1: Information loss $\mathrm{tr}(I_Z - I_A)$ by rule at the design centre (cleaning-council illustration).

loss. Softening quarantine into review yields a large practical gain ($\mathrm{tr}(\Delta) \approx 2.27$); stacking that softening with wider's dual auto labels (`quarantine_wider`) improves further to ≈ 2.16 and is the best non-identity coarse policy here. Collapsing to a binary gate is worst among the listed rules (≈ 3.71).

The core ordering highlighted in prior exploratory work remains:

$$\text{near-identity} \;\leq\; \text{wider} \;\leq\; \text{baseline} \;\leq\; \text{stricter}$$

on *every* cell of the 8×8 design grid (64/64), not only at the centre (numerical tolerances 10^{-9}). Quarantine-to-review inserts *below* wider on the loss axis; `quarantine_wider` inserts still lower and satisfies $\mathrm{tr}(\Delta_{\text{quarantine_wider}}) \leq \mathrm{tr}(\Delta_{\text{quarantine_review}})$ on 65/65 recorded centre-and-grid evaluations. Coarse binary sits *above* stricter—an informative deviation from the original four-rule story.

6.2 Grid behaviour

Figure 2 shows loss heatmaps for five non-identity rules over the parameter grid. Absolute loss varies with θ because both $I_Z(\theta)$ and the mass allocation across f-fibres change, but the qualitative ranking is stable in large regions of the domain: softer, finer action alphabets sit cooler (lower loss) than stricter or binary gates. Figure 3 slices loss against evidence at ambiguity nearest 0.3; near-identity stays at zero, while coarse binary and stricter remain the most lossy.

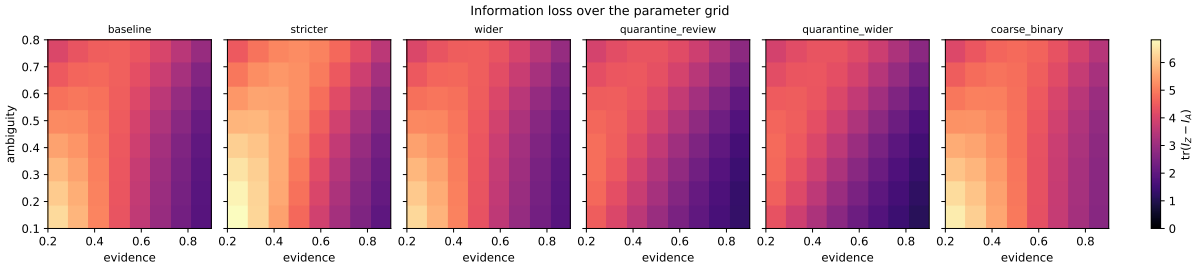


Figure 2: Loss heatmaps over evidence×ambiguity for five coordination rules (cleaning-council illustration).

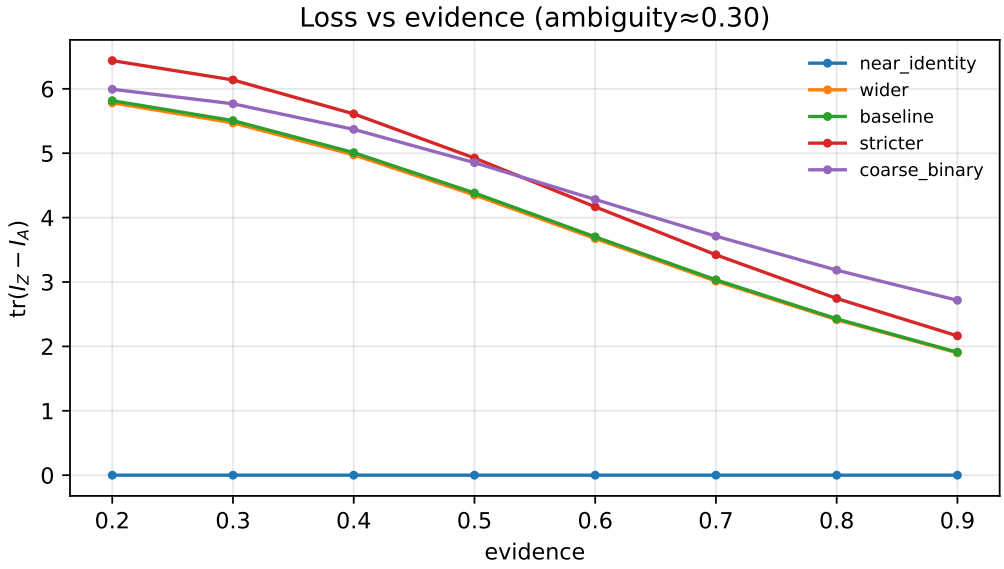


Figure 3: Loss versus evidence at fixed ambiguity near 0.3 (cleaning-council illustration).

6.3 Action masses

Figure 4 shows how probability mass is partitioned at the centre. Stricter reallocates medium-risk mass into reject. Wider splits auto-apply into normalize versus impute. Quarantine-to-review shrinks reject in favour of human review. Coarse binary folds auto-apply and review into `proceed`. These mass shifts are the mechanism behind Fisher loss: merging outcomes with dissimilar score rows collapses distinguishable parameter directions—the same mechanism AggregationMaps will exhibit when merging exchange or quarantine fibres in a sensor field.

6.4 Anisotropy and the quarantine-wider mechanism

Trace summaries hide which parameter direction is discarded. At θ_0, the diagonal entries of Δ_f in the (evidence, ambiguity) basis show that `quarantine_review` cuts ambiguity-axis loss roughly in half relative to baseline ($\Delta_{aa} \approx 0.95$ vs ≈ 1.69) while evidence-axis loss stays comparable. Relative losses $\mathrm{tr}(\Delta)/\mathrm{tr}(I_Z)$ preserve the same ranking where $I_Z(\theta)$ varies across the grid.

Pushforward masses explain the hybrid gain: relative to `quarantine_review`, `quarantine_wider` replaces a single `auto_apply` atom by distinct `auto_normalize` and `auto_impute` atoms and moves additional mass out of `human_review` into `auto_impute`; `reject` mass is unchanged. Finer auto fibres retain score rows that the coarser review rule had merged.

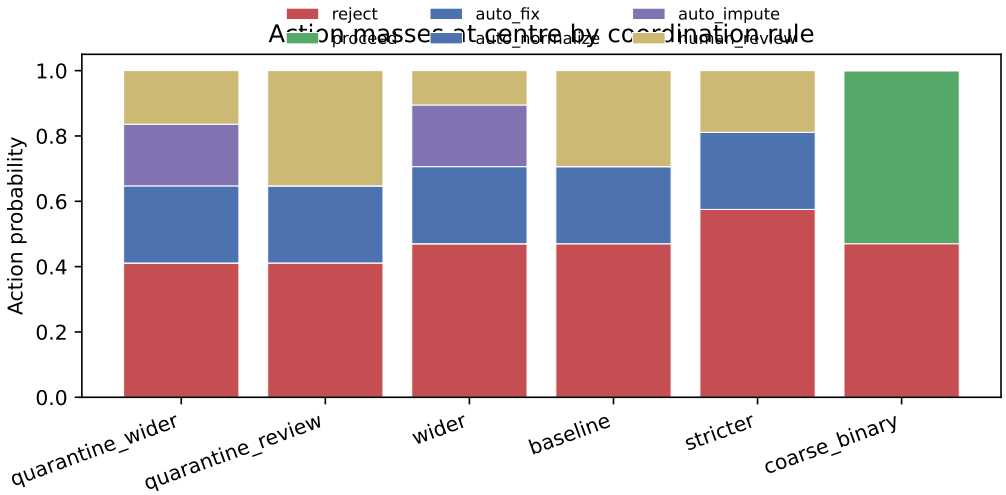


Figure 4: Pushforward action probabilities at the centre for non-identity rules (cleaning-council illustration).

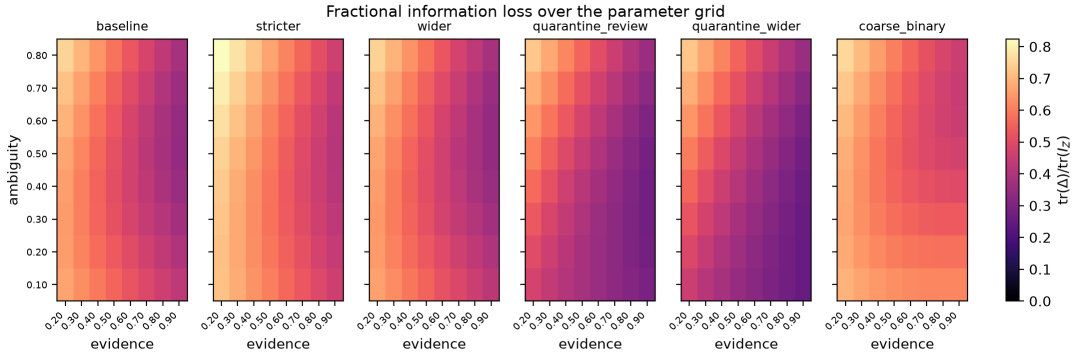


Figure 5: Fractional loss $\mathrm{tr}(\Delta)/\mathrm{tr}(I_Z)$ over the evidence×ambiguity grid for non-identity rules (cleaning-council illustration).

6.5 Sensor-network results

Unit-disk main experiment. We place $N = 9$ motes on a 3×3 grid in $[0,1]^2$ with unit-disk radius $R = 0.45$ (12 edges, mean degree 2.67). The global field parameter is $\theta = (\text{amplitude}, \text{length scale}) = (0.7, 0.35)$ of a Gaussian bump with fixed centre. The same seven AggregationMaps are applied with sensor exchange/quarantine semantics. The Fisher approximation check passes and the core order near-identity $\leq$ wider $\leq$ baseline $\leq$ stricter holds. Among non-trivial rules, `quarantine_wider` has the lowest loss (1.784), ahead of `wider` (1.820) and `baseline` (1.833); `coarse_binary` (2.246) and `stricter` (2.833) lose most (Table 2).

IBRL real-data control. We take a 480-row subsample of the Intel Berkeley Lab labdata[1] over 12 motes (ids 4, 11, 16, 18, 20, 26, 34, 39, 42, 47, 50, 53) selected by farthest-point sampling, with θ the MinMax-normalized mean temperature and humidity and a unit-disk graph at normalized radius 0.35 (6 edges). The Fisher approximation check passes and the core order again holds, but the full ordering differs from unit-disk (Table 3): `near_identity` (0.000) $<$ `wider` (2.784) $\approx$ `baseline` (2.785) $\approx$ `coarse_binary` (2.796) $<$ `quarantine_wider` (3.042) $<$ `quarantine_review` (3.101) $<$ `stricter` (3.474).

[1] https://db.csail.mit.edu/labdata/labdata.html

Table 2: Synthetic unit-disk 2D field ($N = 9$, $R = 0.45$), centre-point traces at $\theta = (0.7, 0.35)$.

Rule	$\mathrm{tr}(I_Z)$	$\mathrm{tr}(I_A)$	$\mathrm{tr}(\Delta)$
`near_identity`	5.788	5.788	0.000
`wider`	5.788	3.969	1.820
`baseline`	5.788	3.955	1.833
`stricter`	5.788	2.955	2.833
`quarantine_review`	5.788	3.913	1.875
`quarantine_wider`	5.788	4.004	1.784
`coarse_binary`	5.788	3.542	2.246

Table 3: IBRL control (12 motes, 480 rows), centre-point traces at the feature means.

Rule	$\mathrm{tr}(I_Z)$	$\mathrm{tr}(I_A)$	$\mathrm{tr}(\Delta)$
`near_identity`	3.943	3.943	0.000
`wider`	3.943	1.159	2.784
`baseline`	3.943	1.158	2.785
`stricter`	3.943	0.469	3.474
`quarantine_review`	3.943	0.842	3.101
`quarantine_wider`	3.943	0.901	3.042
`coarse_binary`	3.943	1.147	2.796

Reading the two tables honestly. On IBRL both quarantine rules lose more than baseline (3.042 and 3.101 versus 2.785)—the reverse of unit-disk, where `quarantine_wider` is the best non-trivial rule. The quarantine safety valve therefore pays an information cost on real data; whether it helps or hurts $\mathrm{tr}(\Delta)$ is data-dependent, and we do not claim a universal ranking beyond the core order. On IBRL, `wider`, `baseline`, and `coarse_binary` are nearly tied (2.78–2.80), so these three rules are practically indistinguishable by this metric on that subnet.

7 Comparison on the IMAS^2 grid-world benchmark

Setup. We reuse the grid world released with IMAS^2 [Shi et al., 2026] (upstream repository `AronYoung414/multi-agent-active-perception-grid-world`, MIT licence, commit `0eae156`; `grid_world_example.py` and `pomdp_grid.py` are vendored unchanged under `third_party/imas2_grid/`). The world is a 10×10 grid with 12 obstacles and 9 candidate sensors. We fix the sensor set to the placement reported by IMAS^2, sensors $2, 3, 4, 6, 8$. The parameter is $\theta =$ (logit of the adversarial prior, slip probability). Each sensor emits a θ-independent adversarial/nominal/uncertain label, and the seven AggregationMaps fuse the five labels. Every rule is evaluated on the same 500 trajectories (seed 0, horizon 14 sensing steps) with an exact Bayes (HMM forward) filter on the fused labels, reporting the conditional entropy H of the agent type and MAP accuracy.

Main result: auditing the aggregation layer with fixed sensors. This experiment does not select sensors; it audits the fusion rule applied on top of a fixed sensor set. Table 4 gives the stochastic, small-range configuration; Table 5 gives $\mathrm{tr}\Delta I$ for all eight fixed/random-orientation configurations. Across all nine configurations (including the retrained-policy rows), the Spearman correlation between the $\mathrm{tr}\Delta I$ ranking and conditional entropy ranges from 0.65 to 0.93, and with negative accuracy from 0.51 to 0.87 (Table 6). Lower Fisher loss thus tracks better downstream inference, though not perfectly. As on IBRL, both quarantine rules lose more than baseline in

Table 4: IMAS^2 grid world, stochastic dynamics, small range, sensors 2-3-4-6-8. Left: fixed orientations; right: reduced retrained IMAS^2 policies (150 iterations/sensor). H = conditional entropy of the type, Acc = MAP accuracy; 500 trajectories, seed 0.

Rule	trΔI	H	Acc	trΔI	H	Acc
`near_identity`	0.0	0.543	0.824	0.0	0.557	0.828
`wider`	115.4	0.676	0.768	92.3	0.677	0.750
`baseline`	118.6	0.675	0.770	93.4	0.685	0.750
`stricter`	138.7	0.687	0.768	113.9	0.704	0.742
`quarantine_review`	140.9	0.809	0.736	137.8	0.791	0.718
`quarantine_wider`	137.7	0.809	0.736	136.7	0.790	0.718
`coarse_binary`	152.7	0.883	0.608	122.8	0.895	0.620

Table 5: trΔI for every orientation policy (F = fixed, R = random) × dynamics (D/S) × range (S/L) configuration.

Rule	F-D-S	F-D-L	F-S-S	F-S-L	R-D-S	R-D-L	R-S-S	R-S-L
`near_identity`	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
`wider`	42.0	212.6	115.4	136.5	21.3	134.3	82.1	149.3
`baseline`	42.0	221.7	118.6	145.5	21.4	136.2	82.7	151.9
`stricter`	98.3	253.5	138.7	150.1	46.4	169.4	96.7	164.5
`quarantine_review`	89.3	292.1	140.9	287.3	56.8	204.0	129.7	224.0
`quarantine_wider`	89.3	283.0	137.7	278.4	56.7	202.2	129.1	221.4
`coarse_binary`	63.8	277.4	152.7	180.8	42.8	176.2	108.8	181.1

every configuration: most cells are labelled "uncertain" at the working point, so quarantining them discards information rather than protecting it.

Secondary: sensor placement (a negative finding). We also asked whether $\sum_t \operatorname{tr} I_Z$ could act as a placement criterion, scoring all 126 five-of-nine placements against a Monte-Carlo MI estimate (Table 7). It cannot be recommended. Under deterministic dynamics the Spearman correlation with MI is 0.60 and the top-5 sets share a single placement; under stochastic dynamics the correlation is -0.15 and the top-5 sets are disjoint. Fisher ranks the IMAS^2 placement 13th (deterministic) and 54th (stochastic). MI enumeration, by contrast, recovers 2-3-4-6-8 as the top placement under deterministic dynamics (and second under stochastic dynamics), which cross-checks our harness against IMAS^2. Fisher scoring is faster (3.6 s and 4.4 s vs. 52.8 s and 52.3 s for MI), but a fast criterion that picks the wrong sensors is not useful, and we do not count speed in its favour. Fisher ΔI is a weak placement criterion and essentially fails under stochastic motion. Our contribution is auditing coordination and aggregation rules, not sensor selection.

IMAS^2 baseline. We re-ran the upstream IMAS^2 training in reduced form: 150 iterations per sensor (the paper uses 1000), in the reported sensor order, taking 2051 s in total; the training entropy after five sensors is 0.558. Separately, Shi et al. [2026] report conditional entropy 0.493 and accuracy 86.0% for IMAS^2 in their stochastic, large-range setting. These reported numbers are not directly comparable with our reduced re-run (different iteration budget, range, and evaluator). The full pipeline on CPU is estimated at over 12 hours and was not run.

Table 6: Spearman correlation of the seven-rule trΔI ranking with conditional entropy and with negative accuracy, per configuration.

Config	$\rho(\Delta I, H)$	$\rho(\Delta I, -\text{Acc})$
F-D-S	0.66	0.51
F-D-L	0.82	0.84
F-S-S	0.93	0.85
F-S-L	0.89	0.86
R-D-S	0.65	0.69
R-D-L	0.82	0.55
R-S-S	0.86	0.84
R-S-L	0.89	0.85
IMAS^2-retrained-S-S	0.89	0.87

Table 7: Fisher vs. MI over all 126 five-sensor placements (small range, fixed orientations). Ranks are of the IMAS^2-reported placement 2-3-4-6-8; times in seconds for scoring all 126.

Dyn.	n	ρ	top-5	Fisher top-1	MI top-1	F-rank	MI-rank	t_F	t_{MI}
deterministic	126	0.60	1	2-4-5-6-8	2-3-4-6-8	13	1	3.6	52.8
stochastic	126	−0.15	0	0-2-3-6-7	1-2-4-6-8	54	2	4.4	52.3

8 Cross-bed AggregationMap transfer: a failed boundary

Protocol. We ask whether a Fisher-ΔI ranking learned on one bed transfers to another *without recalibration*. The three beds are the synthetic unit-disk field, the IBRL control, and the IMAS^2 grid world (fixed-orientation, stochastic dynamics, small range). The same seven AggregationMaps are scored on each bed. Per-bed utilities for unit-disk and IBRL come from `unit_disk_rule_utility.csv` / `ibrl_rule_utility.csv`: a binary y_{share} under the declared joint at θ_{centre}, with exact Bayes conditional entropy H and accuracy. The IMAS^2 utility is the adversary-type τ via the existing HMM filter; the definition therefore differs from the y_{share} utilities and we treat the three beds as *distinct* scoring domains. Heath's transfer criteria are applied to all six directed pairs: *tongyong* (universal) requires $\rho(\text{src}\,\Delta I, \text{tgt}\,\Delta I) \geq 0.5$, $\rho(\text{src}\,\Delta I, \text{tgt}\,H) \geq 0.5$, and $\rho(\text{src}\,\Delta I, -\text{tgt acc}) \geq 0.5$; *zhengxiang* (positive lift) requires the source Fisher pick (excluding `near_identity`) to beat baseline accuracy on the target *and* quarantine to sit in the bottom half by accuracy and by H; S1/S2 are the corresponding permutation and no-contradiction checks; S3 is the already-failed placement negative control from Section 7.

Result: the boundary fails. Overall, tongyong passes on 2/6 pairs, zhengxiang on 0/6, S1 on 0/6, and S2 on 3/6; S3 still fails (Table 8). The ΔI-vs-ΔI ranking transfers somewhat ($\rho = 0.71$–0.79 across all six directions), but Heath's tongyong requires all three Spearmans and mostly fails on ΔI–H or ΔI–accuracy. Zhengxiang never holds: a unit-disk source often selects `quarantine_wider`, which underperforms baseline on the target (e.g. 0.736 vs. 0.770 on IMAS^2; 0.907 vs. 1.000 on IBRL), and quarantine is not stably the worst rule across beds. The two priority directions—unit-disk→IMAS^2 and IMAS^2 →IBRL—both fail. Only `ibrl`→`imas2` and `ibrl`→`unit_disk` pass tongyong, and neither yields positive lift. S1 fails on every pair, typically because $n = 7$ rules produce ties and non-significant permutation p-values.

Framing. This is a failed boundary, not a soft caveat. The evidence in this paper supports *within-bed* AggregationMap auditing only. We do not claim cross-bed generality: a Fisher

Table 8: Cross-bed AggregationMap transfer without recalibration. $\rho(\Delta I, \Delta I)$, $\rho(\Delta I, H)$, $\rho(\Delta I, -\text{acc})$ are Spearman correlations of the seven-rule ranking on the source bed against target ΔI, conditional entropy, and negative accuracy. Tongyong / Zhengxiang / S1 / S2 are the binary Heath criteria (Y/N). Fisher pick is the lowest-ΔI non-`near_identity` rule on the source; pick/base is its accuracy vs. baseline on the target. UD = unit-disk; IMAS^2 = fixed-orientation, stochastic, small-range.

Direction	$\rho_{\Delta I}$	ρ_H	$\rho_{-\text{acc}}$	T	Z	S1	S2	Fisher pick	pick/base
UD→IMAS^2	0.79	0.57	0.45	N	N	N	N	`quarantine_wider`	0.736/0.770
IMAS^2 →UD	0.79	0.45	0.09	N	N	N	Y	`wider`	1.000/1.000
IMAS^2 →IBRL	0.75	0.45	0.11	N	N	N	Y	`wider`	1.000/1.000
IBRL→IMAS^2	0.75	0.64	0.55	Y	N	N	N	`wider`	0.768/0.770
UD→IBRL	0.71	0.47	0.21	N	N	N	N	`quarantine_wider`	0.907/1.000
IBRL→UD	0.71	0.91	0.65	Y	N	N	Y	`wider`	1.000/1.000

ranking that is useful on one bed does not, without recalibration, select a rule that improves utility on another, and the quarantine ranking remains bed-dependent.

9 Discussion

Sensor-network reading. Under layers (i)–(iii), $\text{tr}(\Delta_f)$ is a shared offline score for choosing among exchange and quarantine rules when local agent laws are held fixed. Softer quarantine (delay or human/operator review instead of hard drop) and finer exchange alphabets (distinct neighbour or payload labels) are the direct analogues of the quarantine-wider gains in Table 1. Binary proceed/reject gates remain operationally convenient but informationally costly. The sensor results temper this: quarantine-wider is best on unit-disk but both quarantine rules lose more than baseline on IBRL (Section 6.5). The ranking machinery and DPI certificate transfer across settings; the absolute traces and the position of quarantine in the ranking do not.

What "half the Fisher" means. A 44% drop in $\text{tr}(I_Z)$ after baseline coordination does not mean the system is 44% "wrong." It means that, locally, infinitesimal changes in the parameters move the final action distribution less than they move the joint of agent outputs. If a monitoring dashboard only watches action or exchange rates, it understates how sensitive the underlying agents are. If a downstream estimator tries to invert actions for θ, it faces a strictly harder problem than inverting Z.

Design takeaway. Among implementable three/four-action policies in the cleaning illustration, routing quarantine to review retained more Fisher than baseline or wider alone; combining that softening with distinct auto-normalize/auto-impute labels is strictly better still on this grid. Practitioners can treat $\text{tr}(\Delta_f)$—and, when needed, its parameter-axis diagonals—as a cheap offline score when debating rule changes before A/B tests or firmware flashes.

Relation to Dec-POMDP rewards and IMAS^2. Prediction rewards and negative-entropy objectives [Lauri and Oliehoek, 2020, Lauri et al., 2020] optimize perception policies. IMAS^2 [Shi et al., 2026] further couples agent selection with mutual-information-maximizing sensing policies under Dec-POMDP dynamics and submodular greedy selection. Our metric does not replace those programmes; it audits the *last* deterministic compression after perception (and any selection) is fixed. Selection expands who may observe; Fisher Δ_f ranking asks how much of that observation survives a chosen AggregationMap—including quarantine—at the exchange layer.

API constraints. Many agent and mote platforms expose only sampled outcomes plus deterministic business or firmware rules. Soft, θ-dependent routers would require either a declared conditional action model or a refusal of the DPI certificate—exactly the contract encoded by Jevometry's AggregationMap. Extending the present study to stochastic coordinators is future work and needs additional regularity.

10 Limitations

- The reported empirical numbers come from hand-specified analytic softmax families in the cleaning council, not from fitted live decision models. Absolute traces are illustrative.
- The sensor experiments are small: 9 synthetic motes and a 12-mote, 480-row IBRL subsample at a single centre point. The quarantine ranking flips between them, so rankings beyond the core order are data-dependent and should not be read as universal.
- On the IMAS^2 grid world, $\mathrm{tr}\Delta I$ sums per-step Fisher information and ignores temporal correlation across steps.
- Under deterministic dynamics the slip derivative is a one-sided finite difference at the boundary slip $= 0$.
- The retrained-policy rows replay recorded states and actions and resample observations in our evaluator; they are not fully Bayes-consistent with the upstream training rollouts.
- Cross-bed transfer without recalibration fails Heath's tongyong and zhengxiang criteria on most pairs; $n = 7$ rules yield frequent ties and non-significant permutation tests for S1. Utility definitions also differ across beds (y_{share} on unit-disk/IBRL versus adversary-type τ on IMAS^2), so the failed transfer is a boundary of the present evidence, not a claim about every possible shared utility.
- Conditional independence is declared, not tested.
- We report parameter-axis diagonals and eigenvalues of Δ_f, but full CRLB-style reads would still need an observation likelihood contract the cleaning example does not provide.
- Action alphabets differ across rules; $\mathrm{tr}(I_A)$ comparisons remain meaningful as retained information about θ, but interpretations that treat $|A|$ as utility are out of scope.
- No human, task-reward, or sensing-accuracy study yet validates that lower Fisher loss improves cleaning or field-estimation outcomes.

11 Conclusion

We proposed ranking multi-agent coordination rules by Fisher information loss under a shared declared joint, with decentralized sensor-agent exchange and quarantine as the primary AggregationMap story. Holding local perception fixed, deterministic coordinators can discard a large fraction of the joint Fisher trace; small rule edits (quarantine handling, finer action alphabets, risk or neighbourhood thresholds) move that loss by amounts that are easy to measure offline. The three-agent cleaning council provides a reproducible analytic proof of concept—including the stable ordering near-identity $\leq$ wider $\leq$ baseline $\leq$ stricter and the quarantine-wider gains. On sensor networks the core order holds on both a synthetic unit-disk field and a 12-mote IBRL control; quarantine-wider is best on unit-disk, but on IBRL both quarantine rules lose more than baseline, so the quarantine safety valve pays an information cost on real data and the full ranking is data-dependent. On the IMAS^2 grid world the fusion-rule ranking tracks downstream entropy

and accuracy, but Fisher ΔI is a weak sensor-placement criterion; the method's role is auditing aggregation, not choosing sensors. Cross-bed AggregationMap transfer without recalibration fails as a boundary—tongyong mostly fails, zhengxiang never holds, and the priority directions unit-disk→IMAS^2 and IMAS^2 →IBRL both fail—so the paper's support is for within-bed rule auditing only.

Future work includes (i) scaling the unit-disk and IBRL experiments to more motes, centre points, and time windows, (ii) replacing analytic agents with TypeSafe / live decision-model Jacobians where appropriate, (iii) stochastic coordinators with explicit conditional action models, and (iv) coupling $\mathrm{tr}(\Delta_f)$ with task-level utilities (field estimation error, cleaning quality) in JAAMAS-style empirical studies.

Reproducibility

From the repository root:

```
uv sync --locked
uv pip install matplotlib   # paper figures
uv run python examples/three_agent_cleaning/compare_rules.py
uv run python examples/three_agent_cleaning/paper_experiments.py
```

Artefacts land in `papers/coord-fisher/data` and `figures/`. Cleaning-council tables and figures in this PDF are the current analytic illustration; unit-disk and IBRL artefacts are `data/unit_disk_*` and `data/ibrl_*`; IMAS^2 grid-world artefacts are `data/imas2_*` (`examples/imas2_grid/`). The grid-world code in `third_party/imas2_grid/` is vendored from `AronYoung414/multi-agent-active-perce` (commit `0eae156`) under the MIT licence, with the licence file retained; credit to the IMAS^2 authors [Shi et al., 2026]. Cross-bed transfer artefacts are `data/cross_bed_transfer_*`. This PDF was built with pdflatex from `papers/coord-fisher/main.tex`.

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